MEMORANDUM

To: Arturo Cajal, Department of Mechanical Engineering, Cal Poly SLO

From: Garrett Leighton, gleighto@calpoly.edu; Laura Marie Strobbe, lstrobbe@calpoly.edu; Kelly Graham, kgraha17@calpoly.edu; Gabe Choi, gchoi03@calpoly.edu

Date: March 1, 2026

RE: **Determining Drag Coefficients of Brass and Aluminum Cylinders in Cross Flow using Surface Pressure and Wake Velocity Profile Measurements**

Introduction

In the lab, the drag for a brass and aluminum cylinder in a uniform flow stream was measured using two methods to understand the flow pattern and drag behavior around a cylinder. For the first method, the brass cylinder was placed inside the wind tunnel, and the surface pressure was measured using a pressure tap on the cylinder. For the second method, the wake profile pressures were measured for the brass and aluminum cylinders using a Pitot-static tube. Using these two methods allows for the comparison of the accuracy of each method. The surface pressure method resulted in a drag coefficient of 1.44, while the momentum method resulted in a drag coefficient of 0.70. It was concluded that the surface method yielded a more accurate estimate for the drag on a cylinder because it is more similar to the theoretical value of 1.2 and was less susceptible to inaccuracies.

Background

Understanding fluid behavior around objects such as cylinders is important because it is prevalent in many real-world applications such as the flow around buildings or pillars. Wind tunnels are useful tools for investigating fluid behavior. Specifically, the flow pattern and drag around a cylinder can be determined in a wind tunnel.

Total drag is defined by the sum of the friction drag and pressure drag. The amount of drag on the cylinder and overall flow characteristics are governed by the Reynolds number, shown in Equation 1,

$$Re_D = \frac{UD}{\nu} \tag{1}$$

where U is the free stream velocity, D is the diameter of the cylinder, and ν is the kinematic viscosity of the fluid. For Reynolds number values between $1 * 10^3$ and $2 * 10^5$, also known as the sub-critical range, the flow will have a laminar boundary layer until it reaches the maximum diameter of the cylinder. At this point, the boundary layer will separate from the cylinder and form a large turbulent wake behind the

cylinder. The sub-critical range is characterized by higher drag coefficients, around 1.2 to 1.3, and a substantial amount of pressure drag compared to friction drag.

For Reynolds numbers greater than $2 * 10^5$, called the super-critical range, the boundary layer will transition to turbulent. Because of the increased momentum associated with turbulent flow, the boundary layer will separate from the cylinder at a much later point, yielding a much smaller turbulent wake behind the cylinder. Although the friction drag will increase in the super-critical range due to the turbulent flow, the overall drag coefficient is much lower, typically around 0.3, due to the decrease in pressure drag caused by the reduced size of the turbulent wake. The respective sizes of the turbulent wakes for the sub-critical and super-critical Reynolds number ranges are shown in Figure 1 [1].

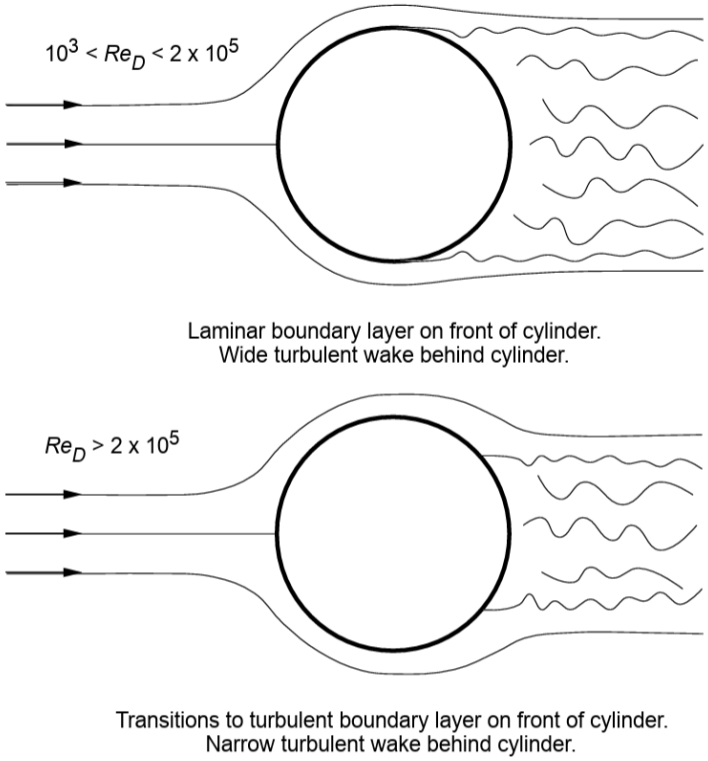


Figure 1. Size of Turbulent Wake for Sub-critical and Super-critical Reynolds Number Ranges.

The dimensionless drag coefficient can be determined experimentally in a wind tunnel. It is especially useful to obtain the dimensionless drag coefficient because it could be used to find the drag force on an object regardless of the fluid, object size, or free stream velocity. Once the dimensionless drag coefficient is obtained, the drag force can be easily obtained by rearranging using Equation 2,

$$C_D = \frac{F_D}{\frac{1}{2}\rho U^2 A}. \quad (2)$$

Procedure/Analysis

For the experiment, the motor speed controller for the wind tunnel was set to 20 Hz. The Pitot-static tube was positioned at x = 5 in and y = 18 in. After steady-state was reached, the pressure reading at this location was recorded and used to determine the free-stream velocity in the wind tunnel. A brass

cylinder with a 1.5-inch diameter was positioned so that its pressure tap was oriented forward in the test section. Using the correct port on the pressure selector, the pressure tap measurements at every 5 degrees were recorded for a full 180-degree rotation of the cylinder. Five measurements were taken at each angle. Only 180 degrees of measurements were obtained because it can be assumed that the pressure on the opposite side of the cylinder would be identical.

Additionally, the Pitot-static tube was used to measure the pressure behind the cylinder. Specifically, the Pitot-static tube was positioned at x = 25 in and y = 8 in. The actual y-axis position was recorded, and five pressure measurements were taken. The Pitot-static tube position was increased 0.5 inches, and five pressure reading were recorded. This process was repeated until the Pitot-static tube reached a y-location of 16 inches.

After all measurements were obtained, the brass cylinder was replaced with an aluminum cylinder with a smaller diameter, 1 inch. Starting from a y-location of 20 inches, a similar procedure was followed to obtain pressure measurements behind the cylinder. Specifically, the y-position of the Pitot-static tube was decreased 0.5 inches until it reached a y-location of 4 inches.

The dimensionless drag coefficient for a cylinder was experimentally determined using two methods: surface pressure method and the momentum method. For the surface pressure method, the drag coefficient is obtained by numerically integrating the dimensionless pressure coefficient. The dimensionless pressure coefficient can be obtained from the surface pressure measurements on the cylinder. The dimensionless pressure coefficient is determined using Equation 3 and the drag coefficient for this method is determined using Equation 4.

$$C_P = \frac{(p - p_o)}{\frac{1}{2}\rho U^2} \tag{3}$$

$$C_D = \int_0^{\pi} C_P \cos\theta \, d\theta \tag{4}$$

where C_D is the dimensionless drag coefficient, F_D is the drag force, and ρ is the density of the fluid.

The drag coefficient can also be determined using a momentum method. For this method, the measured dynamic pressure readings from the Pitot-static tube are used to obtain the velocity at various points behind the streamline. Then, the dimensionless drag coefficient is calculated using Equation 5,

$$C_D = \frac{2}{D}\int_0^{H} \left(1 - \frac{u}{U}\right)\left(\frac{u}{U}\right) dy \tag{5}$$

where u is the velocity at various y-locations along the test section, D is the diameter of the cylinder, and H is the height of the control volume.

Results and Discussion

The free stream velocity was calculated to be 44.9 ft/s using the initial Pitot-static tube measurement. Additionally, using this velocity and the known diameter of the brass cylinder, the Reynolds number associated with the flow around the brass cylinder was calculated to be 3.51×10^4, which belongs in the sub-critical range of Reynolds numbers. The Reynolds number for the aluminum cylinder is 1.76×10^4, which also corresponds to the sub-critical range. These calculations are shown in Attachment A. Additionally, as noted in Attachment A, the free-stream velocity was calculated using the initial Pitot-static tube measurement in the wind tunnel. However, after completing the surface pressure measurements, the Pitot-static tube pressure reading increased, corresponding to a free-stream velocity of 49.0 m/s. As a result, this increased free-stream velocity was used for the wake profile calculations.

The differential pressure measurements were taken using a pressure tap on the brass cylinder. The average differential pressure measurement and statistical uncertainty are shown in Table 1 for a reduced set of cylinder angles. All raw data collected for the surface pressure measurements are shown in Attachment B. The calculated pressure coefficients and uncertainty based on the differential pressure uncertainty are shown in Table 1 as well.

Table 1. Cylinder Angle, Differential Pressure, and Pressure Coefficient with Statistical Uncertainty.

Cylinder Angle, θ (degrees)	Average Differential Pressure Measurement (in H_2O)	Average Pressure Coefficient, C_P
0	0.4514 ± 0.0011	1.0085 ± 0.0025
20	0.2968 ± 0.0010	0.6631 ± 0.0023
40	-0.1592 ± 0.0056	-0.3557 ± 0.0126
60	-0.6170 ± 0.0056	-1.3785 ± 0.0124
80	-0.6768 ± 0.0147	-1.5121± 0.0328
100	-0.6032 ± 0.0130	-1.3477 ± 0.0290
120	-0.6220 ± 0.0089	-1.3897 ± 0.0199
140	-0.6378 ± 0.0154	-1.4250 ± 0.0345
160	-0.6598 ± 0.0247	-1.4741 ± 0.0552
180	-0.7008 ± 0.0172	-1.5657 ± 0.0385

Figure 2 shows the pressure coefficient plotted against cylinder angle. Overall, the trend makes sense. The measured C_p distribution versus cylinder angle shows the expected trend for flow over a circular cylinder. At $\theta = 0°$ (the stagnation point), C_p is near 1, then C_p decreases as the flow accelerates around the cylinder surface. Past about $\theta \approx 30° - 40°$, C_p becomes negative (suction), reaching a minimum around $\theta \approx 70° - 90°$. After this minimum, C_p partially recovers but remains strongly negative and relatively flat from roughly $\theta \approx 100°$ to $180°$. This nearly constant region indicates that the boundary layer has separated because the flow is not ideal and inviscid.

Based on the large suction region and the flat C_p plateau on the rear half of the cylinder, the wake likely consists of separated shear layers forming a broad recirculation region behind the cylinder. This agrees with expectations for our calculated Reynolds number ($\mathrm{Re}_D \approx 3.51 \times 10^4$) as seen in Attachment

A. At this range, a cylinder typically experiences separation on the sides and a wide, low-pressure wake, which produces a strong pressure drag.

The statistical uncertainty in the pressure coefficient was generally small in the attached flow region, indicating highly repeatable measurements. The uncertainty increased significantly in the separated wake region ($\theta \gtrsim 90°$), consistent with the increased unsteadiness of the separated shear layers.

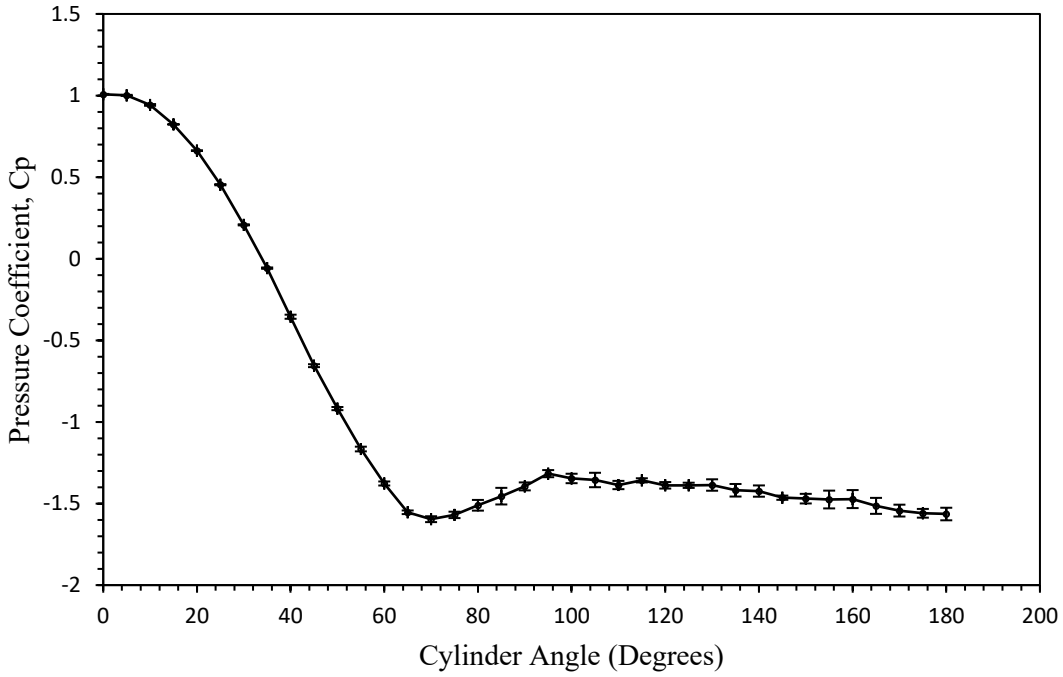


Figure 2. Pressure Coefficient Plotted versus Theta (degrees) For Brass Cylinder with Statistical Uncertainty.

Figure 3 [2] shows the theoretical pressure coefficient plotted against cylinder angle. As shown by the Reynolds number calculation, the flow is around the transition value for turbulent flow, and the experimental pressure coefficient results closely match the curve for laminar flow.

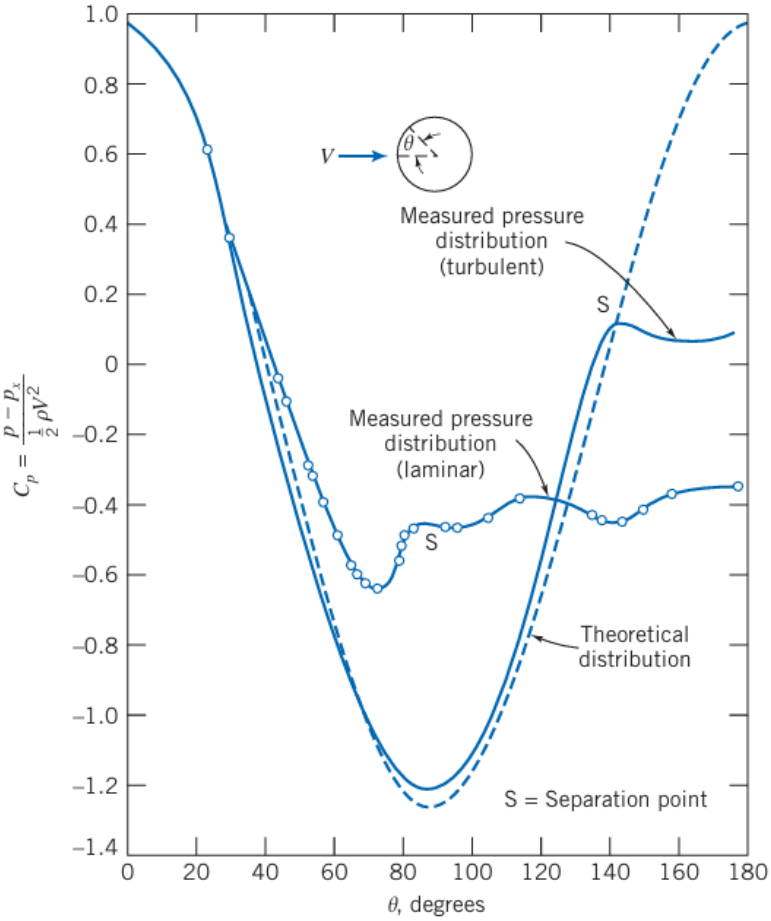


Figure 3. Theoretical Pressure Coefficient versus Cylinder Angle

The streamwise position, vertical position, average Pitot-static tube pressures with statistical uncertainty, and normalized velocity are shown for the brass and aluminum cylinders in Tables 2 and 3, respectively. For the brass cylinder, the velocity decreases as the Pitot-static tube is moved towards the center of the cylinder. As the Pitot-static tube continues to move upwards, the velocity increases again, approaching the free-stream velocity. This trend can be easily seen using the normalized velocity. This matches the theoretical behavior of a fluid over a cylinder. The presence of the cylinder creates the turbulent wake behind the cylinder, and the velocity decreases in this wake.

Table 2. Tabulated Streamwise Position, Vertical position, Normalized Vertical Position (y/D), Mean Pitot-static Probe Pressure, Velocity, and Normalized Velocity (u/U) for Brass Cylinder.

Streamwise Position, x (in)	**Vertical Position, y (in)**	**Normalized Vertical Position, y/D**	**Mean Pitot-Static Pressure (in H_2O)**	**Velocity (ft/s)**	**Normalized Velocity**
25	8.01	-2.66	0.5334 ± 0.0027	49.02	1.000
25	8.52	-2.32	0.5230 ± 0.0030	48.54	0.990
25	9.00	-2.00	0.5086 ± 0.0047	47.86	0.977
25	9.50	-1.67	0.5012 ± 0.0037	47.51	0.970
25	10.01	-1.33	0.4808 ± 0.0079	46.54	0.950
25	10.51	-0.99	0.4424 ± 0.0052	44.64	0.911
25	11.01	-0.66	0.4182 ± 0.0112	43.40	0.886
25	11.50	-0.33	0.3818 ± 0.0135	41.47	0.846
25	12.00	0.00	0.3502 ± 0.0215	39.72	0.810
25	12.52	0.35	0.3550 ± 0.0263	39.99	0.816
25	13.01	0.67	0.4088 ± 0.0165	42.91	0.876
25	13.53	1.02	0.4428 ± 0.0136	44.66	0.911
25	14.01	1.34	0.4652 ± 0.0129	45.77	0.934
25	14.50	1.66	0.4894 ± 0.0115	46.95	0.958
25	15.02	2.01	0.5124 ± 0.0053	48.04	0.980
25	15.51	2.34	0.5186 ± 0.0092	48.33	0.986
25	16.02	2.68	0.5306 ± 0.0034	48.89	0.997

The results for these same parameters for the aluminum cylinder are shown in Table 3. Unlike the brass cylinder, the normalized velocity increases slightly, then decreases again. This does not match the expected behavior of flow over a cylinder. Since the aluminum cylinder has a smaller diameter, the Reynolds number decreases and the flow is more laminar. Because of this, the wake is larger for the smaller aluminum cylinder, which is also why more measurements were needed for the aluminum tube. To cause these issues, the Pitot-static tube may have improperly aligned or positioned incorrectly in the wind tunnel.

Table 3. Tabulated Streamwise Position, Vertical position, Normalized Vertical Position (y/D), Mean Pitot-static Probe Pressure, Velocity, and Normalized Velocity (u/U) for Aluminum Cylinder.

Streamwise Position, x (in)	Vertical Position, y (in)	Normalized Vertical Position, y/D	Mean Pitot-Static Pressure (in H_2O)	Velocity (ft/s)	Normalized Velocity
25	20.00	8.00	1.0170 ± 0.0020	67.68	1.381
25	19.50	7.50	1.0160 ± 0.0009	67.65	1.380
25	18.99	6.99	1.0184 ± 0.0014	67.73	1.382
25	18.50	6.50	1.0190 ± 0.0020	67.75	1.382
25	18.01	6.01	1.0196 ± 0.0014	67.77	1.383
25	17.50	5.50	1.0214 ± 0.0014	67.83	1.384
25	16.99	4.99	1.0228 ± 0.0016	67.87	1.385
25	16.55	4.55	1.0218 ± 0.0022	67.84	1.384
25	16.01	4.01	1.0212 ± 0.0063	67.82	1.384
25	15.50	3.50	1.0236 ± 0.0014	67.90	1.385
25	14.97	2.97	1.0254 ± 0.0014	67.96	1.387
25	14.47	2.47	1.0284 ± 0.0014	68.06	1.389
25	13.95	1.95	1.0340 ± 0.0020	68.24	1.392
25	13.46	1.46	1.0398 ± 0.0024	68.44	1.396
25	12.94	0.94	1.0482 ± 0.0024	68.71	1.402
25	12.45	0.45	1.0500 ± 0.0025	68.77	1.403
25	11.95	-0.05	1.0512 ± 0.0039	68.81	1.404
25	11.50	-0.50	1.0494 ± 0.0034	68.75	1.403
25	10.95	-1.05	1.0428 ± 0.0016	68.53	1.398
25	10.45	-1.55	1.0384 ± 0.0026	68.39	1.395
25	9.95	-2.05	1.0326 ± 0.0021	68.20	1.392
25	9.48	-2.52	1.0272 ± 0.0010	68.02	1.388
25	9.02	-2.98	1.0252 ± 0.0018	67.95	1.387
25	8.47	-3.53	1.0262 ± 0.0018	67.99	1.387
25	7.96	-4.04	1.0240 ± 0.0009	67.91	1.386
25	7.48	-4.52	1.0250 ± 0.0023	67.95	1.386
25	6.94	-5.06	1.0250 ± 0.0012	67.95	1.386
25	6.46	-5.54	1.0250 ± 0.0009	67.95	1.386
25	5.99	-6.01	1.0234 ± 0.0011	67.89	1.385
25	5.46	-6.54	1.0236 ± 0.0014	67.90	1.385
25	4.98	-7.02	1.0280 ± 0.0009	68.05	1.388
25	4.48	-7.52	1.0268 ± 0.0010	68.01	1.388
25	3.97	-8.03	1.0270 ± 0.0015	68.01	1.388

Figure 4 shows the normalized velocity versus the normalized position for the brass and aluminum cylinders. As discussed previously, the brass cylinder follows the expected velocity profile for a cylinder in cross-flow. The normalized wake velocity profile shows a clear momentum deficit centered near y/D=0, with the minimum velocity occurring near the cylinder centerline and the velocity recovering to

approximately $\frac{u}{U} \approx 1$ at larger |y/D|. This behavior is consistent with the formation of a separated wake behind the cylinder, where viscous mixing and turbulence gradually restore the freestream velocity away from the centerline of the wake. However, for the aluminum cylinder, these trends are not followed. This could have been caused by a problem with the setup. Specifically, more accurate results could have been obtained if the Pitot-static tube was positioned further at an increased distance away from the cylinder in the streamwise direction. This would have caused the Pitot-static tube to be exposed to more one-dimensional laminar flow. A Pitot-static tube is not recommended for measuring flow that is not one-dimensional. The turbulent wake has flow in three dimensions, so it cannot accurately measure the pressure well.

The pressure, and thus velocity, was only measured at one stream-wise location, but the wake will increase in size as the streamline direction is increased. As the streamwise direction is increased, the velocity will increase as well because the fluid recovers from the significant loss of momentum caused by the cylinder.

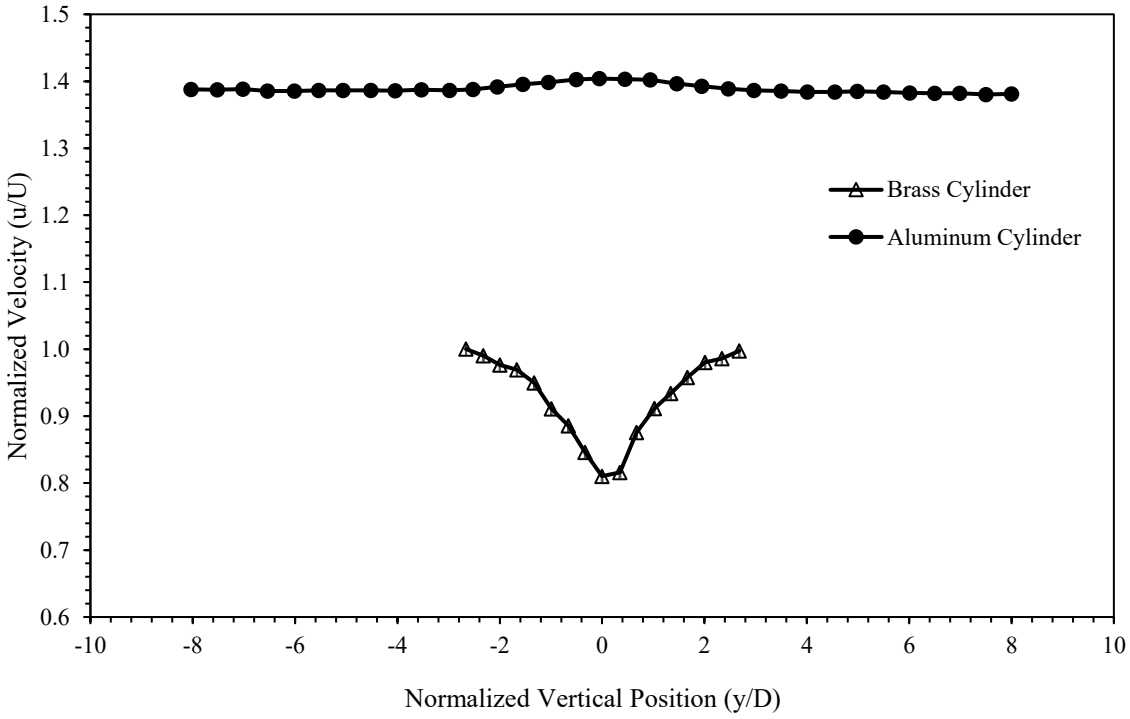


Figure 4. Normalized Velocity versus Normalized Vertical Position.

The drag coefficient for the surface pressure method was calculated using Equation 4 and the pressure coefficients shown in Table 1. The resulting drag coefficient is 1.44 for the brass cylinder. Sample calculations are shown in Attachment C. For the momentum method, Equation 5 and the normalized velocity values from Table 2 were used to obtain the drag coefficient. The resulting drag coefficient for this method is 0.70. Sample calculations for the momentum method are also shown in Attachment C. The calculated drag coefficients for the brass cylinder are tabulated in Table 4, along with the theoretical value for the drag coefficient. The theoretical value for the drag coefficient was obtained from Figure 5 [1].

Table 4. Theoretical and Experimental Drag Coefficients for Brass Cylinder

Method		Theoretical Value	Reynolds Number
Surface Pressure	**Momentum**		
1.44	0.70	1.2	3.51×10^5

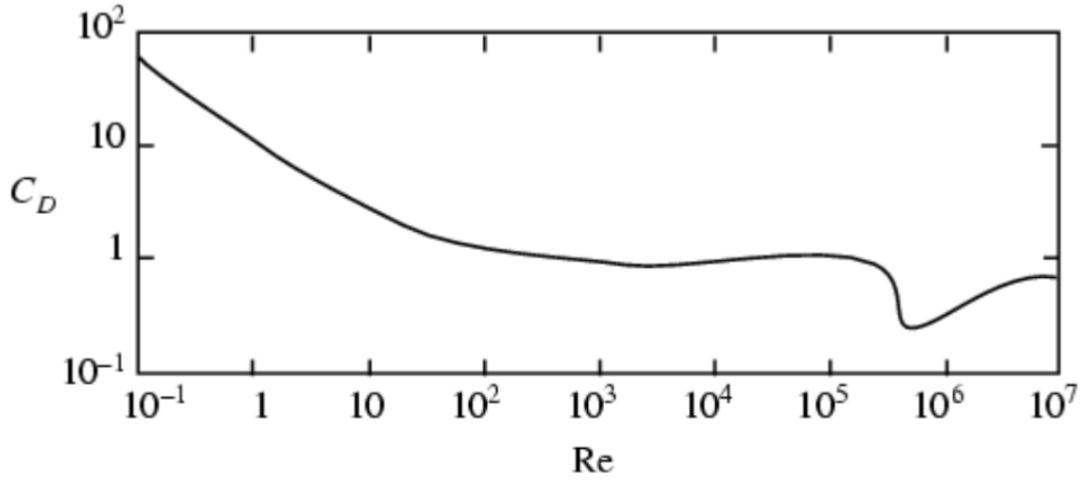


Figure 5. Theoretical Drag Coefficient versus Reynolds number

The theoretical value for the drag coefficient is approximately 1.2. This value was also discussed in the Background section of the report; this is a typical value for the drag coefficient in the sub-critical range. Comparing the experimental values, the surface pressure method yields a more accurate measurement of the drag coefficient. This could be explained by the fact that all of the wake profile measurements were made using Pitot-static tube. Pitot-static tubes are not well-suited to measure pressure for three-dimensional flow. Since the Pitot-static tube is being intentionally placed in the wake, the flow is turbulent, which is not one-dimensional. To obtain more accurate measurements for the momentum method, the Pitot-static tube could have been placed farther from the cylinder in the streamwise direction. This is because the flow would be closer to laminar and one-dimensional if it is farther away from the cylinder, theoretically yielding more accurate measurements.

For the aluminum tube, the drag coefficient was calculated using the momentum method shown in Attachment C. However, because the velocity values were deemed inaccurate, the drag coefficient result is unreasonable. Specifically, the drag coefficient was calculated to be 17.3. Since the diameter is smaller for the aluminum tube, it would be reasonable that the drag coefficient is higher because it is more laminar flow with a smaller relative wake size. However, this value is much higher than its theoretical value, as Figure 5 indicates that the drag coefficient should still fall at about 1.2 for this smaller Reynolds number.

When visualizing the flow around the brass and aluminum cylinders, we would expect that the boundary layer would separate from the aluminum cylinder earlier than that of the brass cylinder. This is because the Reynolds number is smaller due to the smaller diameter, meaning that the flow will have less momentum. This lack of momentum compared to a turbulent flow means that the boundary layer cannot withstand the adverse pressure gradient on the cylinder, leading to boundary layer separation at an earlier point. Because of this, the turbulent wake will be larger.

References

1. ME 347 – Lab 7 Manual, Arturo Cajal.
2. Fox and McDonald's Introduction to Fluid Mechanics, 9th Edition, Philip J. Pritchard and John W. Mitchell.

Attachments

A. Calculation: Free-Stream Velocity and Reynolds number
B. Cylinder Angle, Differential Pressure, and Pressure Coefficient
C. Sample Calculations: Drag Coefficients for Surface Pressure and Momentum Method

ENGINEERING DATA SHEET	Sheet No.	A – 1/3	Date: 02/28/26
Attachment A: Calculations for Free-Stream Velocity and Reynolds Number	Prepared	GC	02/28/26
	Checked	KG	02/28/26
	Approved	LMS	02/28/26

Given (from data sheet)

- $p_{amb} = 14.627\ psia$
- $T_{amb} = 70\ °F$
- $p_{zero} = -0.008\ inH_2O$
- $p_{\infty} = 0.44\ inH_2O$
- Brass cylinder diameter: D = 1.50 in
- $R_{air} = 1716\ \frac{ft \cdot lbf}{slug \cdot °R}$
- $1\ inH_2O = 5.202\ psf$
- $\nu_{air}(\approx 70°F) \approx 1.6 \times 10^{-4}\ \frac{ft^2}{s}$

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1) Correct the Pitot dynamic pressure

Equation:

$$q_{\infty} = p_{\infty} - p_{zero}$$

Calc:

$$q_{\infty} = 0.44 - (-0.008) = 0.448\ inH_2O$$

Convert to psf:

$$q_{\infty}(psf) = q_{\infty}(inH_2O) \times 5.202 = 0.448 \times 5.202 = 2.33\ psf$$

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2) Compute air density from ambient conditions

Convert pressure:

$$p_{amb}(psf) = p_{amb}(psia) \times 144 = 14.627 \times 144 = 2106\ psf$$

Convert temperature:

$$T_{amb}(°R) = T_amb(°F) + 459.67 = 70 + 459.67 = 529.67\ °R$$

Ideal gas:

$$\rho = \frac{p}{R\,T}$$

Calc:

$$\rho = \frac{2106}{1716 \times 529.67} = 0.00231 \frac{slug}{ft^3}$$

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3) Compute freestream velocity from Pitot relation

Pitot dynamic pressure relation:

$$q_\infty = (1/2)\,\rho\,{U_\infty}^2$$

Solve:

$$U_\infty = \sqrt{2\frac{q_\infty}{\rho}}$$

Calc:

$$U_\infty = \sqrt{2 \times \frac{2.33}{0.00231}} = \sqrt{2017} = 44.9 \frac{ft}{s}$$

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4) Reynolds number based on cylinder diameter (brass)

Convert diameter:

$$D(ft) = \frac{D(in)}{12} = \frac{1.50}{12} = 0.125\ ft$$

Reynolds number:

$$Re_D = U_\infty \frac{D}{\nu}$$

Calc:

$$Re_D = \frac{44.9 \times 0.125}{1.6 \times 10^{-4}} = 3.51 \times 10^4$$

Reynolds number based on cylinder diameter (aluminum)

Convert diameter:

$$D(ft) = \frac{D(in)}{12} = \frac{1.00}{12} = 0.083\ ft$$

Reynolds number:

$$Re_D = U_\infty \frac{D}{\nu}$$

Calculate:

$$Re_D = \frac{44.9\ \frac{ft}{s} \times 0.083\ ft}{1.6 \times 10^{-4}} = 1.76 \times 10^4$$

Note that the free-stream velocity used for these calculations is 44.9 ft/s, which was based on the initial Pitot-static tube reading for the experiment. We noted that the Pitot-static reading changed when collecting data for the wake profile. The free-stream velocity used for this portion of the experiment was 49.0 ft/s, matching the first measurement for the Pitot-static tube during the wake profile measurements. This free-stream velocity corresponds to a Reynolds number of 4.08×10^4 for the brass cylinder and 2.04×10^4 for the aluminum cylinder.

ENGINEERING DATA SHEET	Sheet No.	B – 1/3	Date: 02/28/26
Attachment B: Cylinder Angle, Differential Pressure, and Pressure Coefficient	Prepared	GC	02/28/26
	Checked	KG	02/28/26
	Approved	GL	02/28/26

Brass Cylinder		
Angle	P (in H2O)	Cp
0	0.452	1.009
	0.451	1.007
	0.452	1.009
	0.452	1.009
	0.45	1.004
5	0.448	1.000
	0.447	0.998
	0.448	1.000
	0.45	1.004
	0.449	1.002
10	0.425	0.949
	0.423	0.944
	0.422	0.942
	0.42	0.938
	0.421	0.940
15	0.369	0.824
	0.37	0.826
	0.368	0.821
	0.368	0.821
	0.369	0.824
20	0.297	0.663
	0.296	0.661
	0.296	0.661
	0.297	0.663
	0.298	0.665
25	0.205	0.458
	0.204	0.455
	0.202	0.451
	0.203	0.453
	0.202	0.451
30	0.093	0.208
	0.095	0.212
	0.092	0.205
	0.092	0.205
95	-0.597	-1.333
	-0.587	-1.310
	-0.58	-1.295
	-0.599	-1.337
	-0.587	-1.310
100	-0.614	-1.371
	-0.588	-1.313
	-0.602	-1.344
	-0.612	-1.366
	-0.6	-1.339
105	-0.601	-1.342
	-0.621	-1.386
	-0.626	-1.397
	-0.602	-1.344
	-0.587	-1.310
110	-0.624	-1.393
	-0.612	-1.366
	-0.611	-1.364
	-0.63	-1.406
	-0.629	-1.404
115	-0.612	-1.366
	-0.611	-1.364
	-0.605	-1.350
	-0.612	-1.366
	-0.602	-1.344
120	-0.633	-1.413
	-0.618	-1.379
	-0.624	-1.393
	-0.621	-1.386
	-0.614	-1.371
125	-0.619	-1.382
	-0.623	-1.391
	-0.629	-1.404
	-0.625	-1.395
	-0.614	-1.371
130	-0.624	-1.393
	-0.627	-1.400

		0.093	0.208				-0.614	-1.371	
		-0.026	-0.058				-0.637	-1.422	
		-0.026	-0.058				-0.604	-1.348	
	35	-0.028	-0.063				-0.645	-1.440	
		-0.025	-0.056				-0.615	-1.373	
		-0.024	-0.054			135	-0.636	-1.420	
		-0.162	-0.362				-0.631	-1.408	
		-0.164	-0.366				-0.651	-1.453	
	40	-0.159	-0.355				-0.647	-1.444	
		-0.152	-0.339				-0.65	-1.451	
		-0.159	-0.355			140	-0.643	-1.435	
		-0.295	-0.658				-0.623	-1.391	
		-0.29	-0.647				-0.626	-1.397	
	45	-0.294	-0.656				-0.659	-1.471	
		-0.291	-0.650				-0.657	-1.467	
		-0.298	-0.665			145	-0.654	-1.460	
		-0.416	-0.929				-0.649	-1.449	
		-0.407	-0.908				-0.661	-1.475	
	50	-0.413	-0.922				-0.67	-1.496	
		-0.412	-0.920				-0.657	-1.467	
		-0.409	-0.913			150	-0.659	-1.471	
		-0.525	-1.172				-0.642	-1.433	
		-0.525	-1.172				-0.665	-1.484	
	55	-0.527	-1.176				-0.652	-1.455	
		-0.514	-1.147				-0.664	-1.482	
		-0.521	-1.163			155	-0.632	-1.411	
		-0.623	-1.391				-0.68	-1.518	
		-0.615	-1.373				-0.677	-1.511	
	60	-0.611	-1.364				-0.666	-1.487	
		-0.619	-1.382				-0.625	-1.395	
		-0.617	-1.377			160	-0.672	-1.500	
		-0.695	-1.551				-0.663	-1.480	
		-0.694	-1.549				-0.673	-1.502	
	65	-0.695	-1.551				-0.68	-1.518	
		-0.693	-1.547				-0.674	-1.504	
		-0.703	-1.569			165	-0.683	-1.525	
		-0.722	-1.612				-0.653	-1.458	
		-0.71	-1.585				-0.702	-1.567	
	70	-0.722	-1.612				-0.706	-1.576	
		-0.712	-1.589				-0.673	-1.502	
		-0.709	-1.583			170	-0.685	-1.529	
	75	-0.711	-1.587				-0.692	-1.545	
		-0.692	-1.545				-0.701	-1.565	

		-0.706	-1.576			175	-0.702	-1.567
		-0.702	-1.567				-0.686	-1.531
		-0.705	-1.574				-0.7	-1.563
	80	-0.68	-1.518				-0.712	-1.589
		-0.685	-1.529				-0.694	-1.549
		-0.683	-1.525			180	-0.715	-1.596
		-0.656	-1.464				-0.686	-1.531
		-0.68	-1.518				-0.691	-1.542
	85	-0.651	-1.453				-0.696	-1.554
		-0.682	-1.522				-0.716	-1.598
		-0.651	-1.453					
		-0.634	-1.415					
		-0.641	-1.431					
	90	-0.638	-1.424					
		-0.629	-1.404					
		-0.622	-1.388					
		-0.615	-1.373					
		-0.621	-1.386					

ENGINEERING DATA SHEET	Sheet No.	C – 1/1	Date: 02/28/26
Attachment C: Sample Calculations: Drag Coefficients for Surface Pressure and Momentum Method	Prepared	KG	02/28/26
	Checked	LS	02/28/26
	Approved	GL	02/28/26

Drag Coefficient Sample Calculations:

Surface Pressure Method:

$$C_D = \int_0^{\pi} C_p \cos\theta \, d\theta$$

Numerically integrate experimental data using trapezoidal rule:

$$\frac{1}{2}\left[f(x_i) + f(x_{i+1})\right](x_{i+1} - x_i)$$

Where $f(x)$ is $C_p \cos\theta$ and x_i is the cylinder angle.

Plug in C_p and angles for the first sum:

$$S_1 = \frac{1}{2}\left[C_p \cos 0^\circ + C_p \cos 5^\circ\right]\left(5^\circ\left(\frac{\pi}{180}\right) - 0^\circ\left(\frac{\pi}{180}\right)\right)$$

Substituting calculated values for C_p:

$$S_1 = \frac{1}{2}\left[1.009 \cos 0^\circ + 1.002 \cos 5^\circ\right](0.087 - 0)$$

$$S_1 = 0.08755$$

This process is repeated between each subsequent angle measurement. Adding up all of these values will lead to a calculated drag coefficient of $\boxed{1.44}$.

The values used for these calculations are shown below.

SURFACE PRESSURE METHOD CALCULATIONS					
theta_deg	theta_rad	d_theta	Cp	f = Cp*cos(theta)	Partial Sum
0	0.000	0.087	1.009	1.009	
5	0.087	0.087	1.002	0.998	0.088
10	0.175	0.087	0.943	0.929	0.084
15	0.262	0.087	0.824	0.796	0.075
20	0.349	0.087	0.663	0.623	0.062
25	0.436	0.087	0.454	0.411	0.045
30	0.524	0.087	0.208	0.180	0.026
35	0.611	0.087	-0.058	-0.047	0.006
40	0.698	0.087	-0.356	-0.272	-0.014
45	0.785	0.087	-0.656	-0.464	-0.032
50	0.873	0.087	-0.919	-0.591	-0.046
55	0.960	0.087	-1.167	-0.669	-0.055
60	1.047	0.087	-1.379	-0.689	-0.059
65	1.134	0.087	-1.555	-0.657	-0.059
70	1.222	0.087	-1.597	-0.546	-0.053
75	1.309	0.087	-1.571	-0.407	-0.042
80	1.396	0.087	-1.512	-0.263	-0.029
85	1.484	0.087	-1.456	-0.127	-0.017
90	1.571	0.087	-1.396	0.000	-0.006
95	1.658	0.087	-1.318	0.115	0.005
100	1.745	0.087	-1.348	0.234	0.015
105	1.833	0.087	-1.357	0.351	0.026
110	1.920	0.087	-1.388	0.475	0.036
115	2.007	0.087	-1.359	0.574	0.046
120	2.094	0.087	-1.390	0.695	0.055
125	2.182	0.087	-1.390	0.797	0.065
130	2.269	0.087	-1.388	0.892	0.074
135	2.356	0.087	-1.420	1.004	0.083
140	2.443	0.087	-1.425	1.092	0.091
145	2.531	0.087	-1.466	1.201	0.100
150	2.618	0.087	-1.471	1.274	0.108
155	2.705	0.087	-1.477	1.338	0.114
160	2.793	0.087	-1.474	1.385	0.119
165	2.880	0.087	-1.516	1.464	0.124
170	2.967	0.087	-1.545	1.521	0.130
175	3.054	0.087	-1.561	1.555	0.134
180	3.142	0.087	-1.566	1.566	0.136
				TOTAL:	1.44

Momentum Method:

Drag Coefficient definition using momentum method:

$$C_D = \frac{2}{D}\int_0^H \left(1 - \frac{V_2}{U}\right)\left(\frac{V_2}{U}\right) dy$$

Note: V_2 is the same as "u". The lab manual uses V_2.

Similar to surface pressure method, the trapezoid method is used:

$$\frac{1}{2}\left[f(x_i) + f(x_{i+1})\right](x_{i+1} - x_i)$$

Substituting $f(x) = \left(1 - \frac{u}{U}\right)\left(\frac{u}{U}\right)$ and x_i = y location

$$S_1 = \frac{1}{2}\left(\left(1 - \frac{u_1}{U}\right)\left(\frac{u_1}{U}\right) + \left(1 - \frac{u_2}{U}\right)\left(\frac{u_2}{U}\right)\right)(y_2 - y_1)$$

$$S_1 = \frac{1}{2}(-0.00011 + 0.00959)(0.71ft - 0.67ft)$$

$$S_1 = 0.000201$$

All of these terms were added. The sum was multiplied by $\frac{2}{D}$.

Sum = 0.044

$\boxed{C_D = 0.70}$

Values used for the calculation are shown below.

x (in)	y (in)	y (ft)	Mean Pressure (in H2O)	u (ft/s)	u/U	Integrand (1-u/U)*(u/U)	Partial Sum
25	8.01	0.67	0.533	49.02	1.000	0.000	
25	8.52	0.71	0.523	48.54	0.990	0.010	0.0002
25	9.00	0.75	0.509	47.86	0.977	0.023	0.0006
25	9.50	0.79	0.501	47.51	0.969	0.030	0.0011
25	10.01	0.83	0.481	46.54	0.950	0.048	0.0016
25	10.51	0.88	0.442	44.64	0.911	0.081	0.0027
25	11.01	0.92	0.418	43.40	0.886	0.101	0.0038
25	11.50	0.96	0.382	41.47	0.846	0.130	0.0047
25	12.00	1.00	0.350	39.72	0.810	0.154	0.0059
25	12.52	1.04	0.355	39.99	0.816	0.150	0.0066
25	13.01	1.08	0.409	42.91	0.876	0.109	0.0053
25	13.53	1.13	0.443	44.66	0.911	0.081	0.0041
25	14.01	1.17	0.465	45.77	0.934	0.062	0.0029
25	14.50	1.21	0.489	46.95	0.958	0.040	0.0021
25	15.02	1.25	0.512	48.04	0.980	0.019	0.0013
25	15.51	1.29	0.519	48.33	0.986	0.014	0.0007
25	16.02	1.34	0.531	48.89	0.997	0.003	0.0003
						Sum	0.043958
						Cd	0.703325

The same approach was followed for the aluminum cylinder.